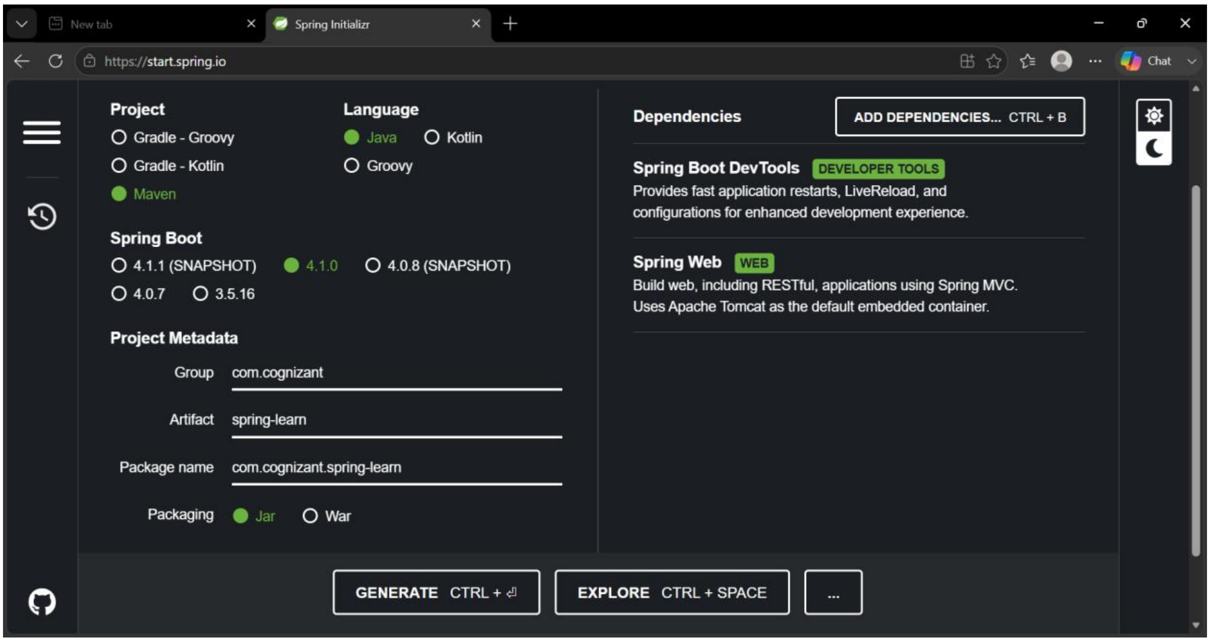
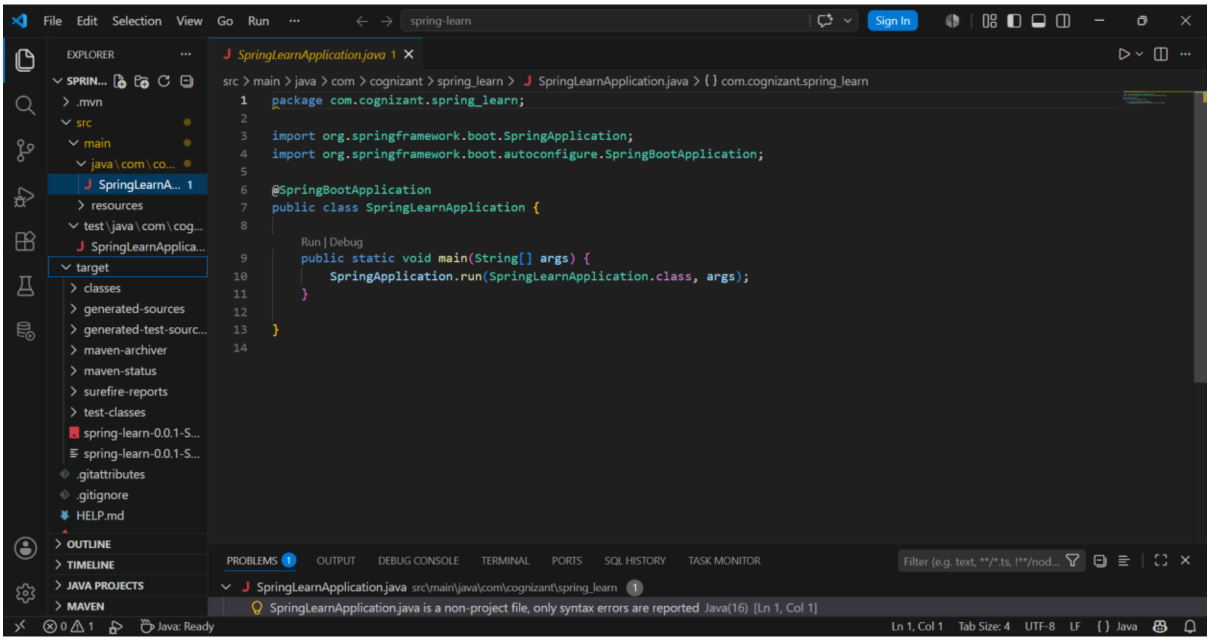
DIGITAL NURTURE 5.0 DEEPSKILLING

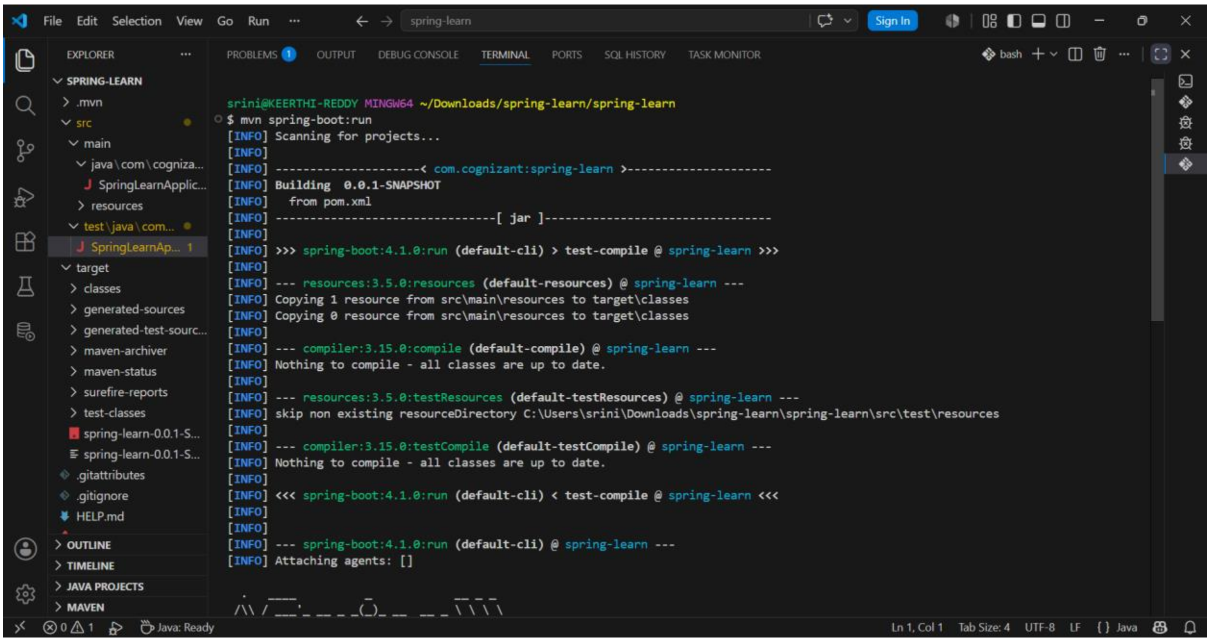
WEEK 3 – HANDS-ON

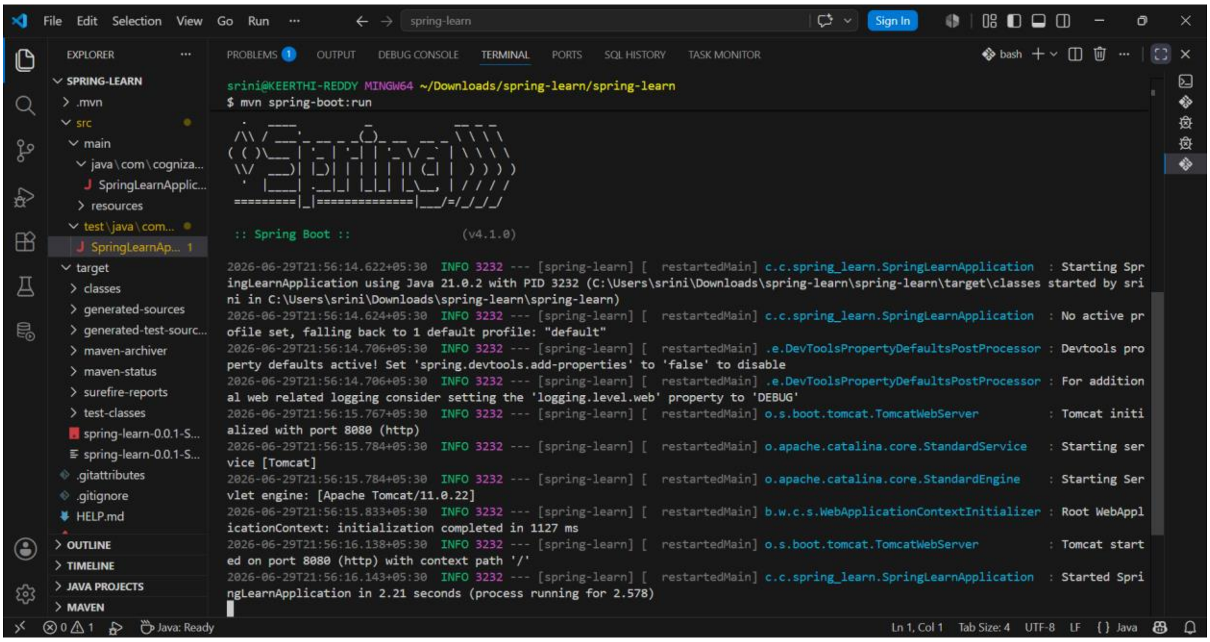
Spring REST using Spring Boot 3

Create a Spring Web Project using Maven

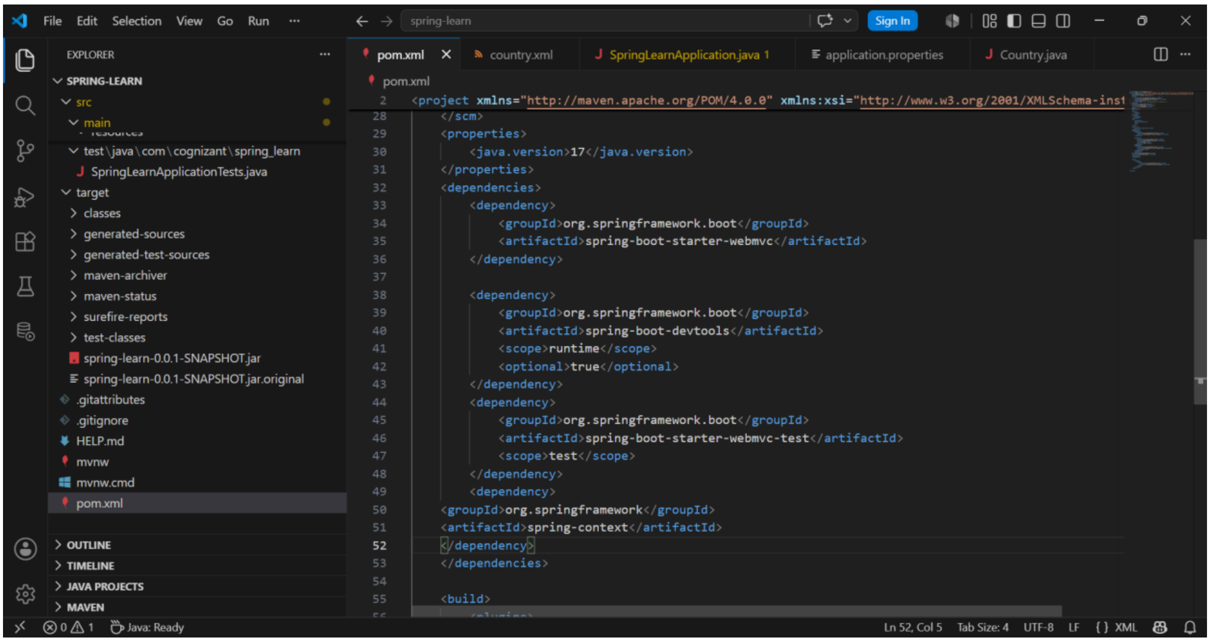


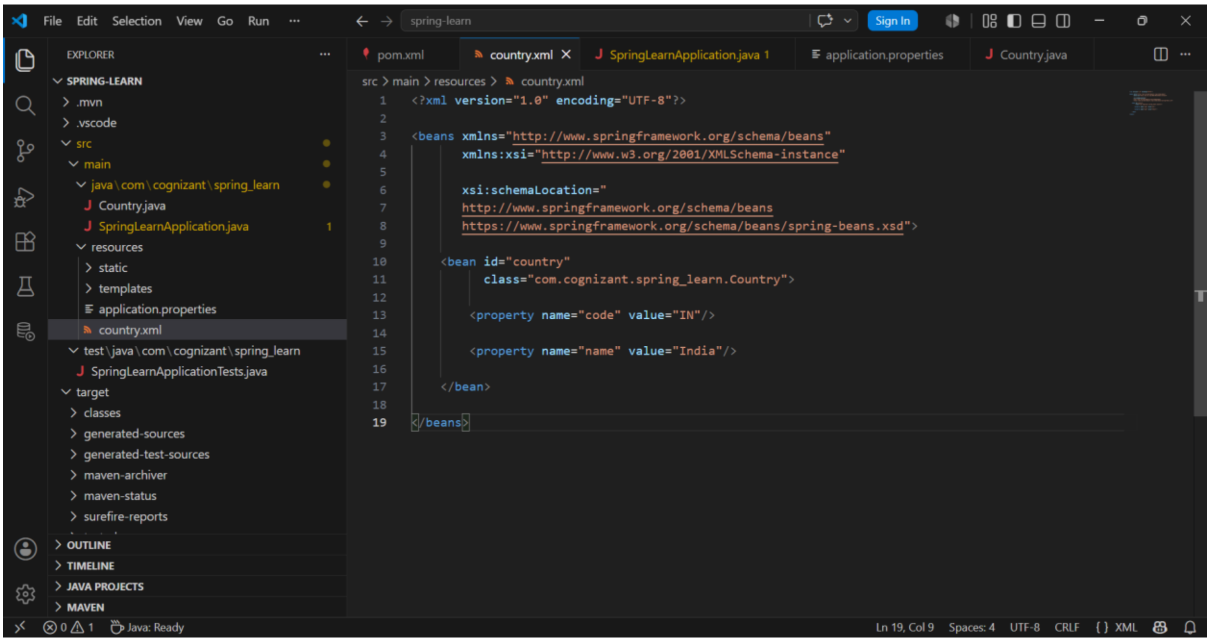


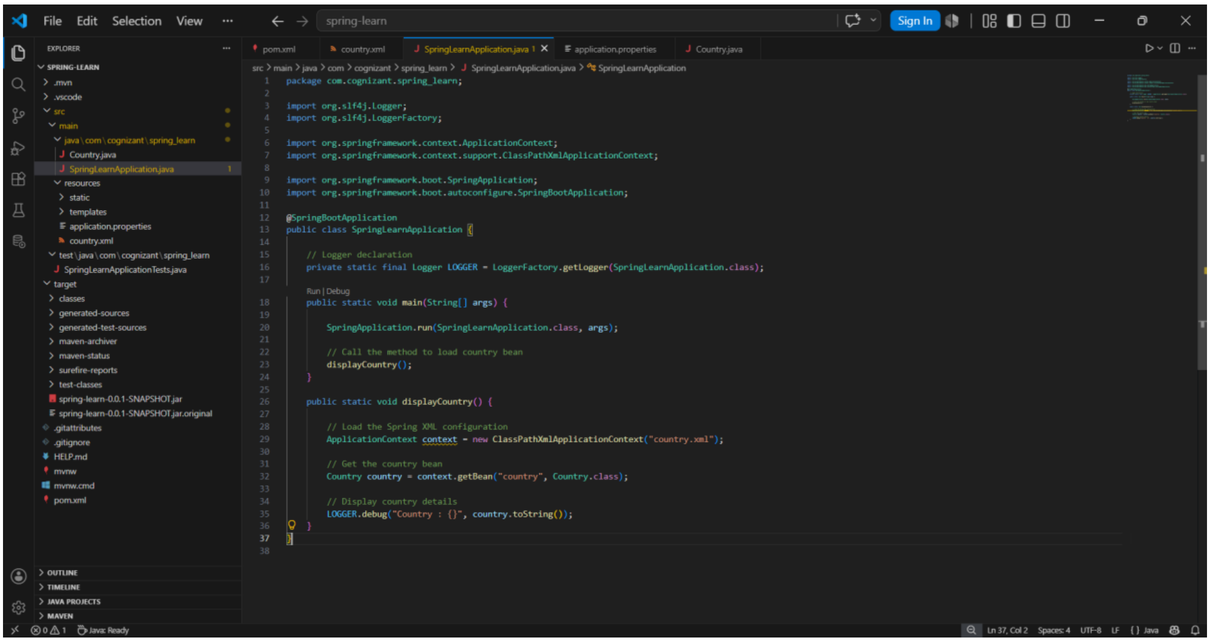

srini@KEERTHI-REDDY MINGW64 ~/Downloads/spring-learn/spring-learn
$ mvn spring-boot:run
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.cognizant:spring-learn >--------------------
[INFO] Building 0.0.1-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[jar]---------------------------------
[INFO]
[INFO] >>> spring-boot:4.1.0:run (default-cli) > test-compile @ spring-learn >>>
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ spring-learn ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ spring-learn ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ spring-learn ---
[INFO] skip non existing resourceDirectory C:\Users\srini\Downloads\spring-learn\spring-learn\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ spring-learn ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] <<< spring-boot:4.1.0:run (default-cli) < test-compile @ spring-learn <<<
[INFO]
[INFO]
[INFO] --- spring-boot:4.1.0:run (default-cli) @ spring-learn ---
[INFO] Attaching agents: []

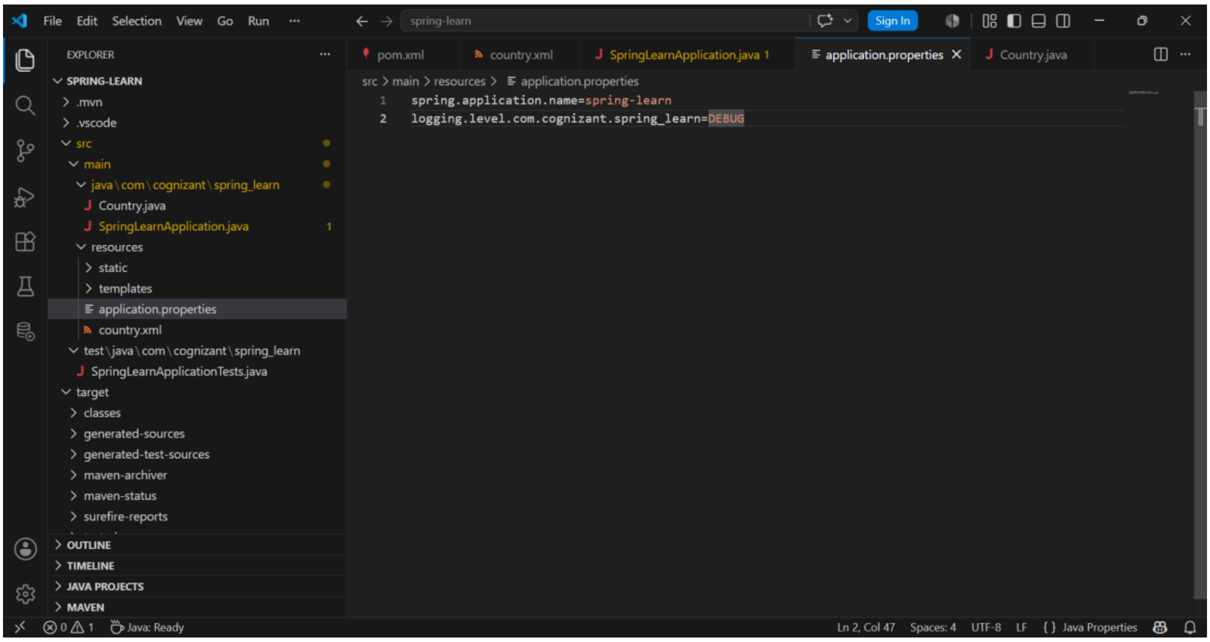

srini@KEERTHI-REDDY MINGW64 ~/Downloads/spring-learn/spring-learn
$ mvn spring-boot:run
:: Spring Boot ::   (v4.1.0)
2026-06-29T21:56:14.622+05:30 INFO 3232 --- [spring-learn] [restartedMain] c.c.spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 21.0.2 with PID 3232 (C:\Users\srini\Downloads\spring-learn\spring-learn\target\classes started by srini in C:\Users\srini\Downloads\spring-learn\spring-learn)
2026-06-29T21:56:14.624+05:30 INFO 3232 --- [spring-learn] [restartedMain] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-06-29T21:56:14.706+05:30 INFO 3232 --- [spring-learn] [restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
2026-06-29T21:56:14.706+05:30 INFO 3232 --- [spring-learn] [restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property to 'DEBUG'
2026-06-29T21:56:15.767+05:30 INFO 3232 --- [spring-learn] [restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-06-29T21:56:15.784+05:30 INFO 3232 --- [spring-learn] [restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-06-29T21:56:15.784+05:30 INFO 3232 --- [spring-learn] [restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-06-29T21:56:15.833+05:30 INFO 3232 --- [spring-learn] [restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 1127 ms
2026-06-29T21:56:16.138+05:30 INFO 3232 --- [spring-learn] [restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-06-29T21:56:16.143+05:30 INFO 3232 --- [spring-learn] [restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 2.21 seconds (process running for 2.578)

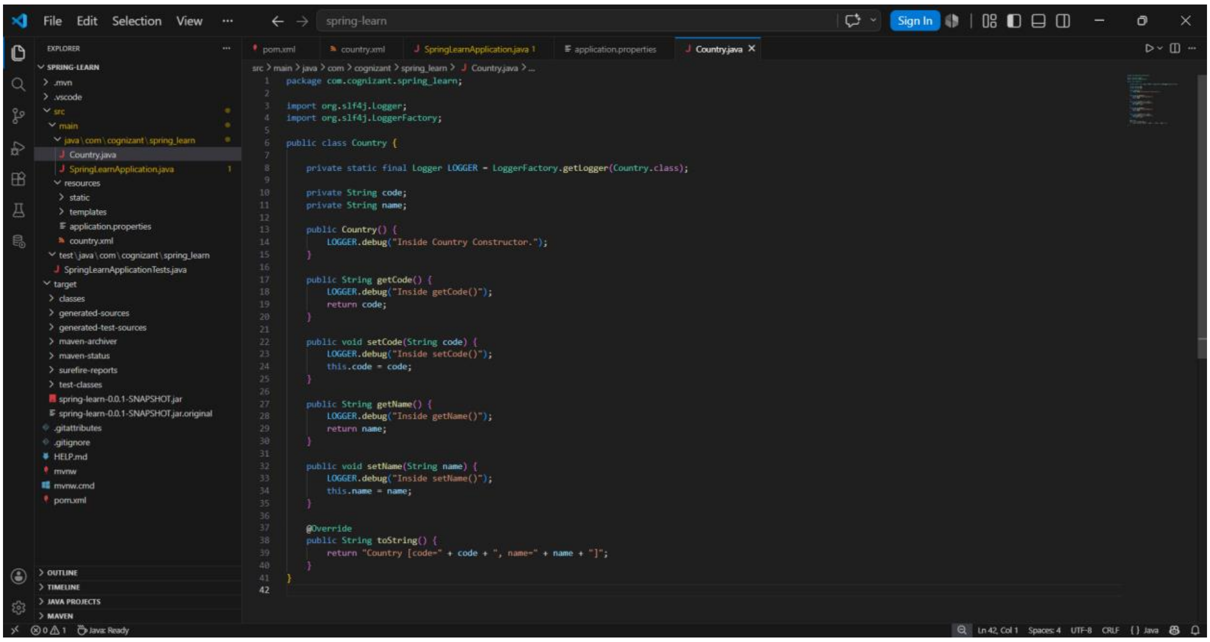
Spring Core – Load Country from Spring Configuration XML

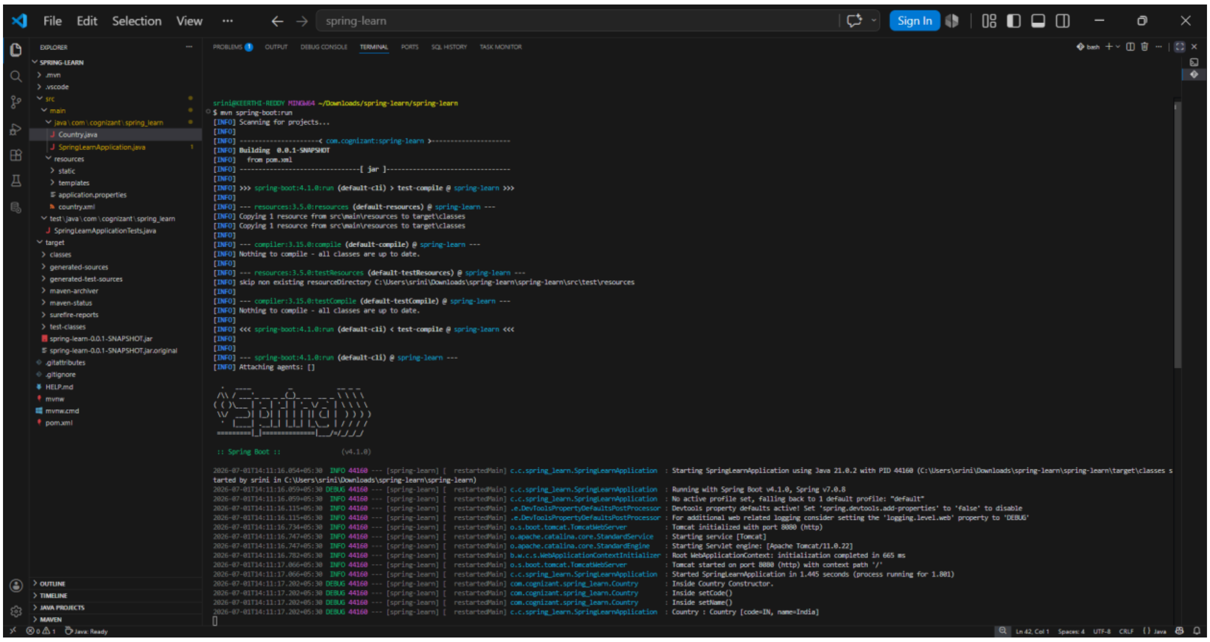




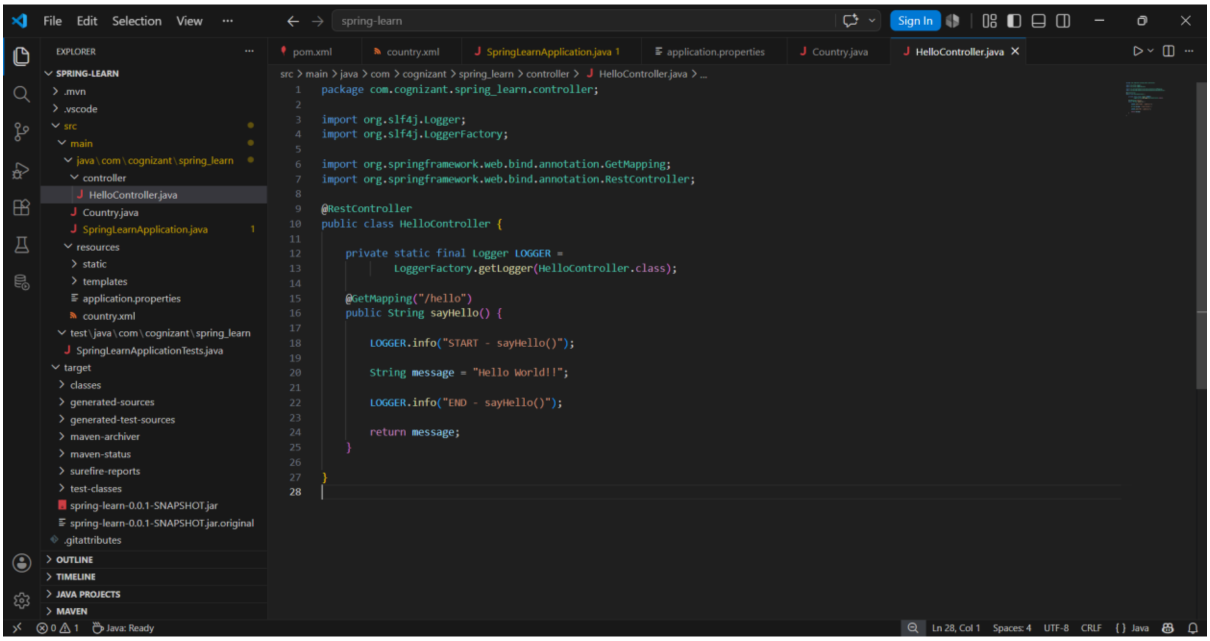

spring-learn
pom.xml
country.xml
SpringLearnApplication.java
application.properties
Country.java
src > main > java > com > cognizant > spring_learn > SpringLearnApplication.java > SpringLearnApplication
package com.cognizant.spring_learn;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
@SpringBootApplication
public class SpringLearnApplication {
// Logger declaration
private static final Logger LOGGER = LoggerFactory.getLogger(SpringLearnApplication.class);
Run | Debug
public static void main(String[] args) {
SpringApplication.run(SpringLearnApplication.class, args);
// Call the method to load country bean
displayCountry();
}
public static void displayCountry() {
// Load the Spring XML configuration
ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
// Get the country bean
Country country = context.getBean("country", Country.class);
// Display country details
LOGGER.debug("Country : {}", country.toString());
}
}
Ln 37, Col 2
Spaces: 4
UTF-8
LF
Java

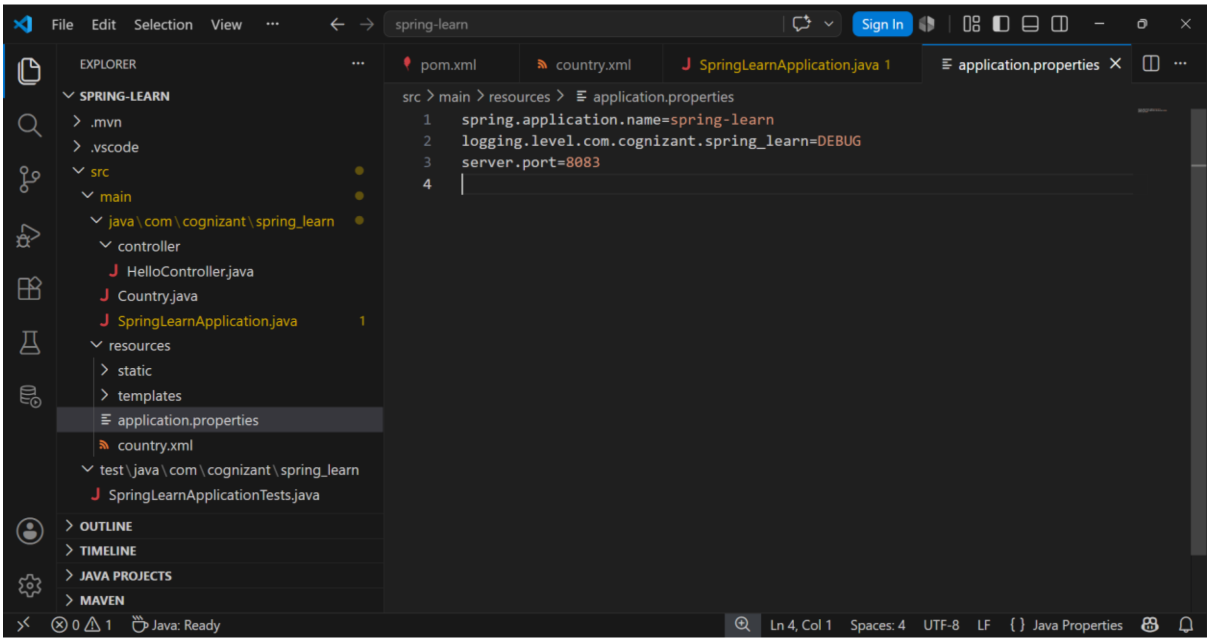

spring-learn
EXPLORER
SPRING-LEARN
pom.xml
country.xml
SpringLearnApplication.java
application.properties
Country.java
src > main > resources > application.properties
spring.application.name=spring-learn
logging.level.com.cognizant.spring_learn=DEBUG
Ln 2, Col 47
Spaces: 4
UTF-8
LF
Java Properties
Java: Ready

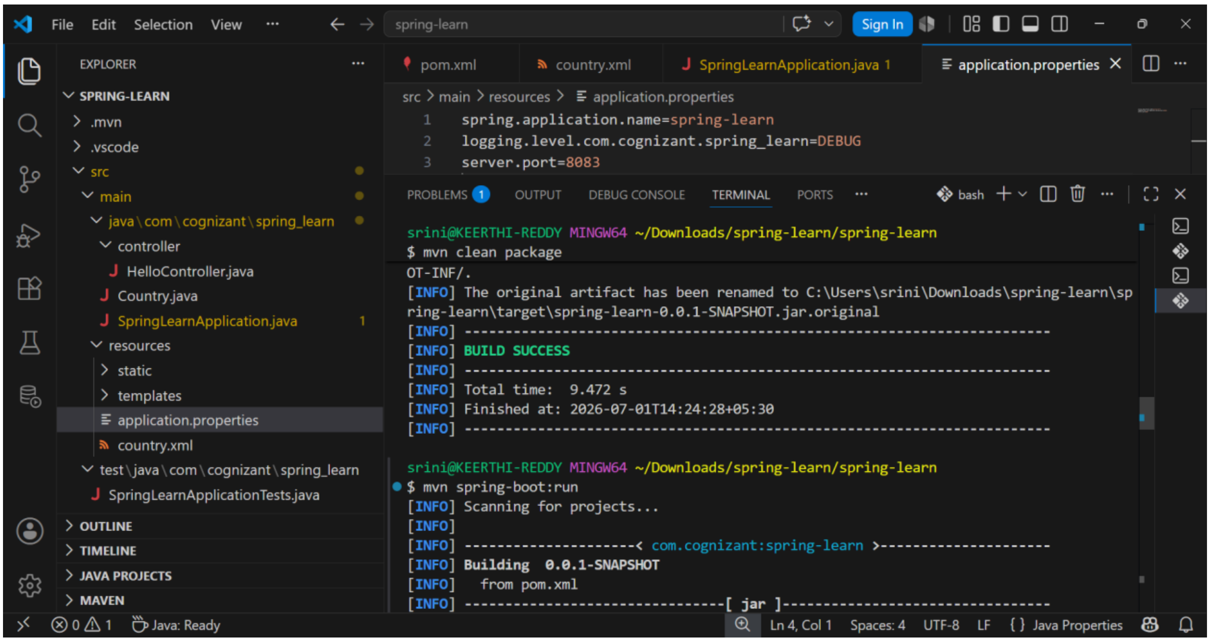




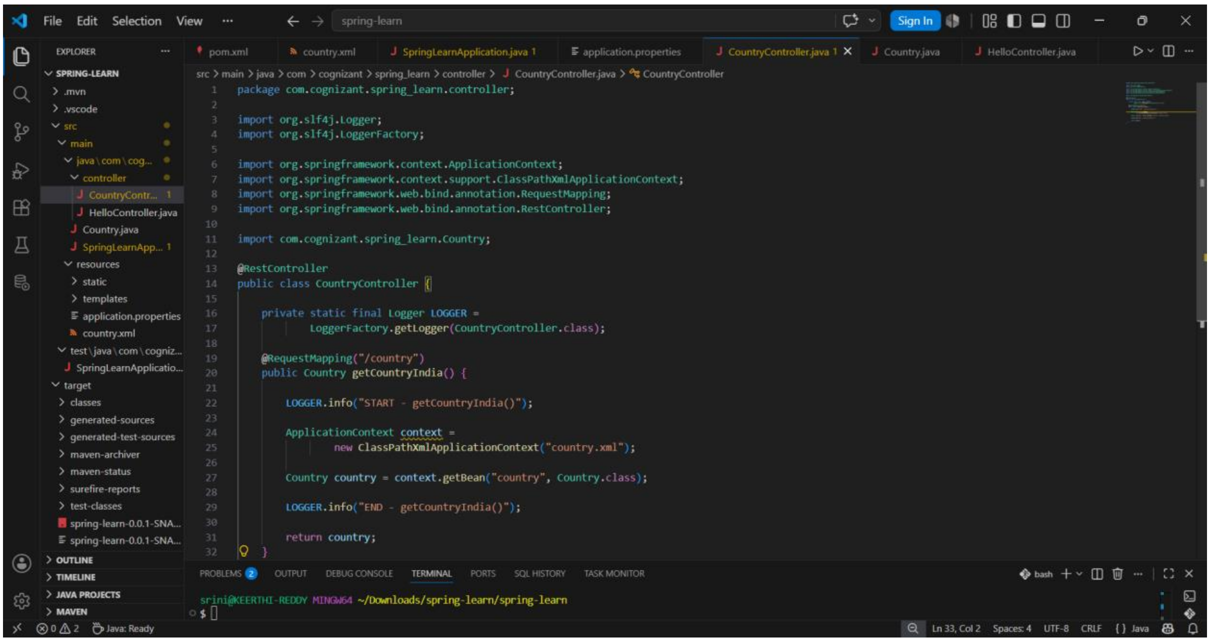
Hello World RESTful Web Service

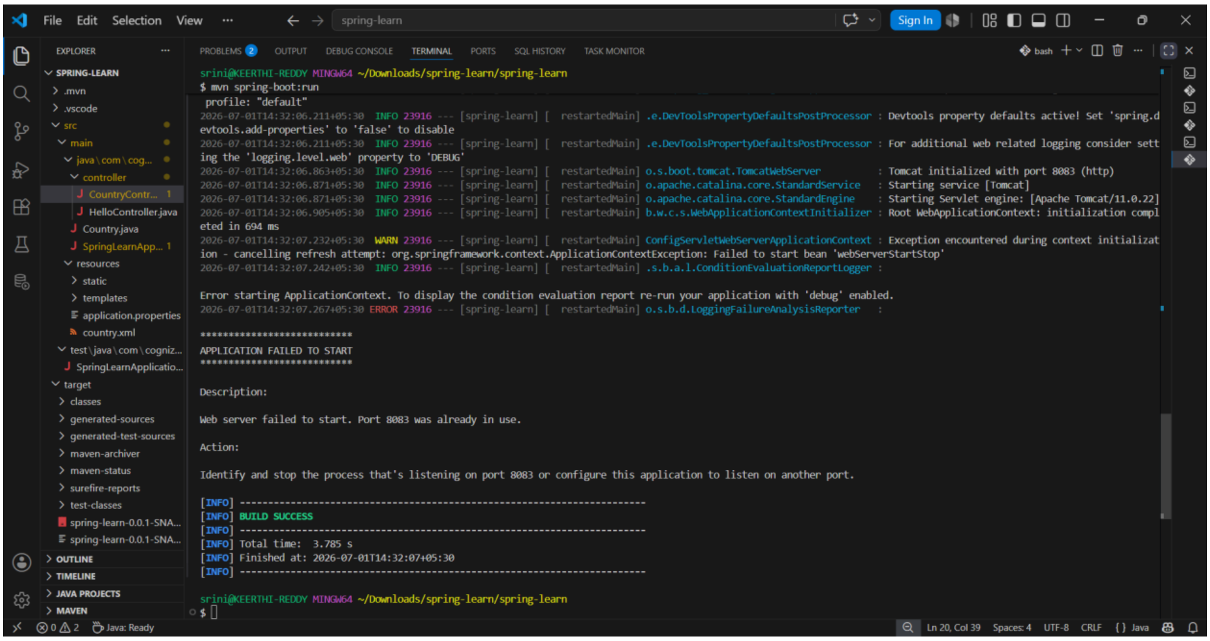

spring-learn
EXPLORER
SPRING-LEARN
HelloController.java
Country.java
SpringLearnApplication.java
application.properties
country.xml
SpringLearnApplicationTests.java
src > main > java > com > cognizant > spring_learn > controller > HelloController.java > ...
package com.cognizant.spring_learn.controller;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;
@RestController
public class HelloController {
private static final Logger LOGGER =
LoggerFactory.getLogger(HelloController.class);
@GetMapping("/hello")
public String sayHello() {
LOGGER.info("START - sayHello()");
String message = "Hello World!!";
LOGGER.info("END - sayHello()");
return message;
}
}
Ln 28, Col 1
Spaces: 4
UTF-8
CRLF
Java
Java: Ready

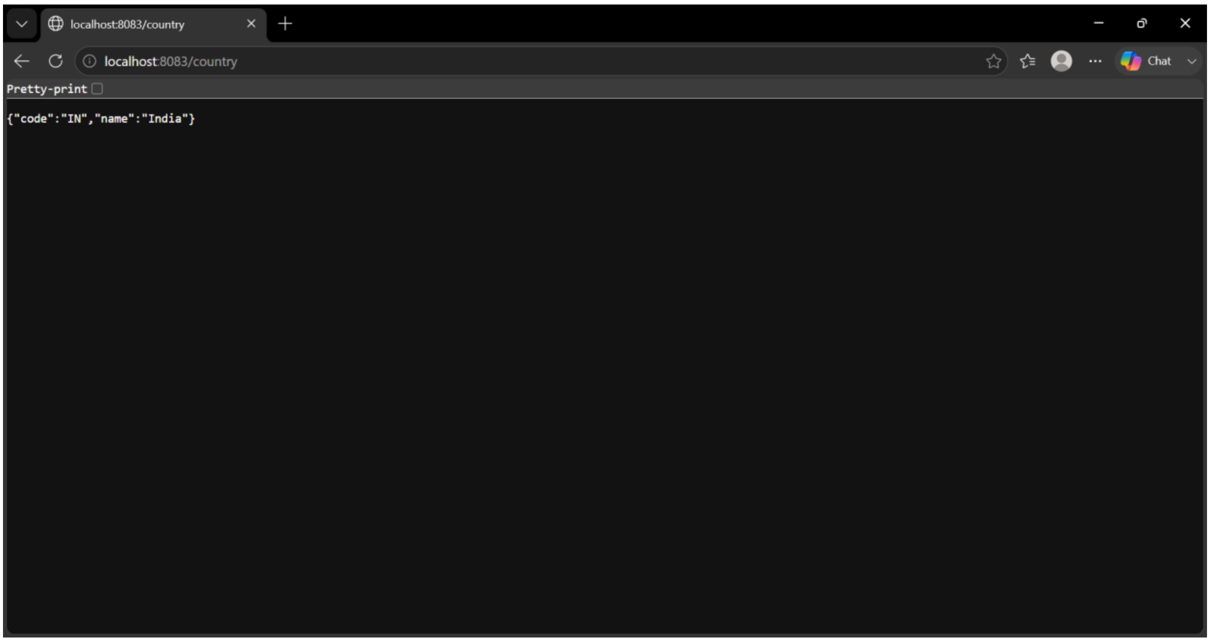

spring-learn
EXPLORER
SPRING-LEARN
src > main > resources > application.properties
spring.application.name=spring-learn
logging.level.com.cognizant.spring_learn=DEBUG
server.port=8083
Ln 4, Col 1
Spaces: 4
UTF-8
LF
Java Properties
Java: Ready



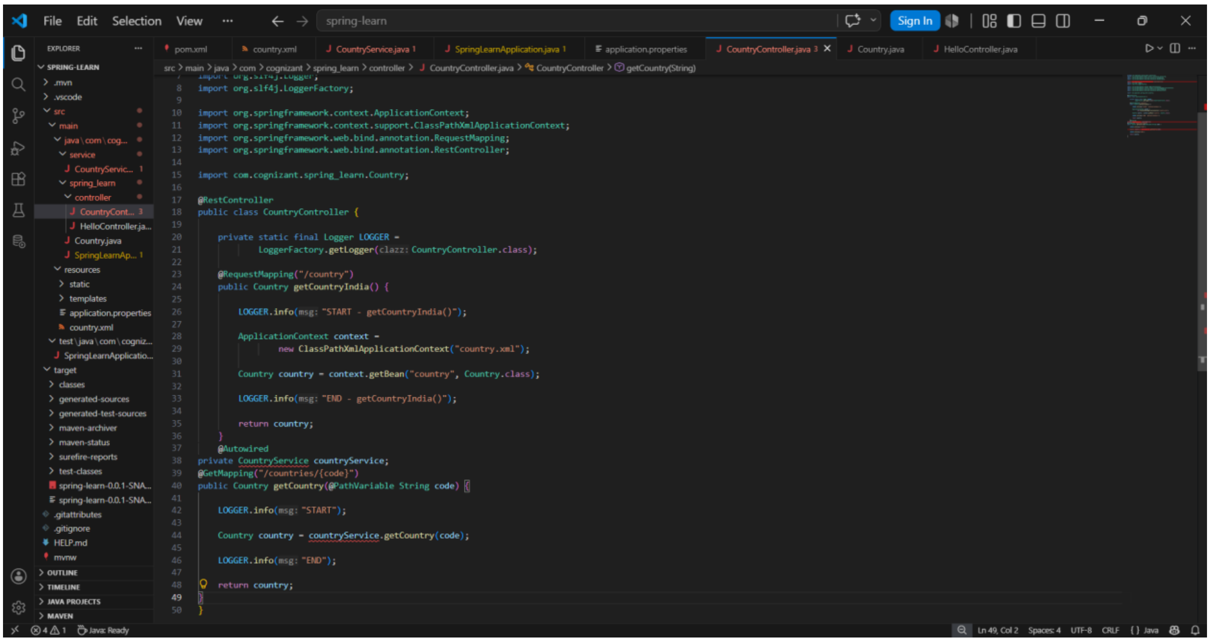
REST - Country Web Service

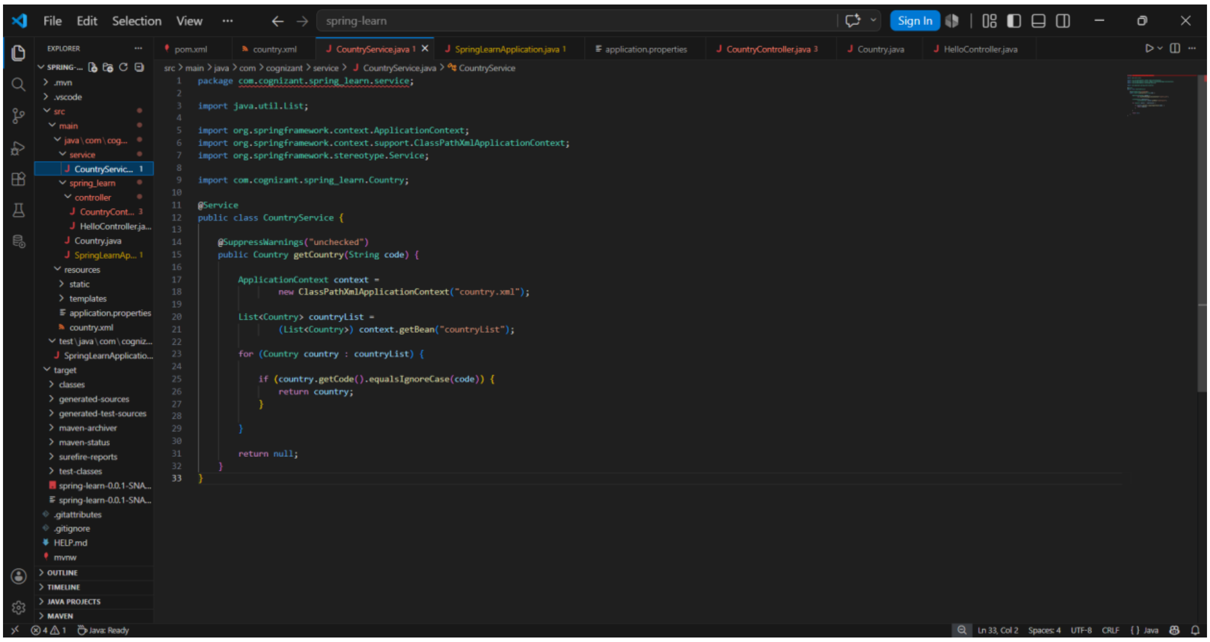


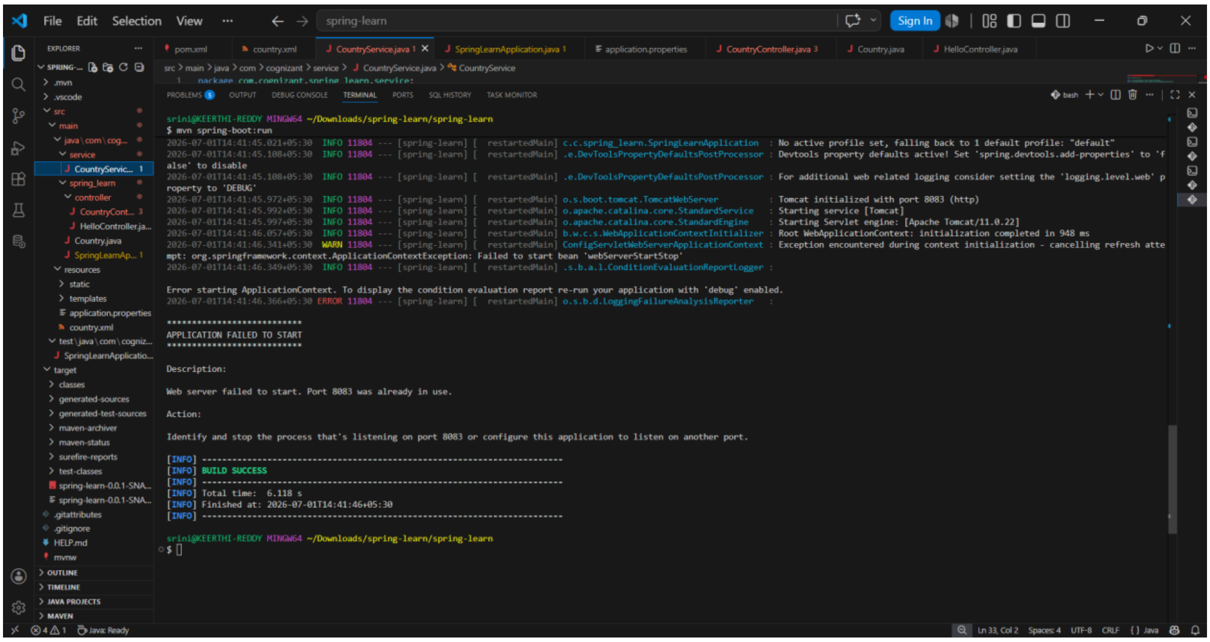


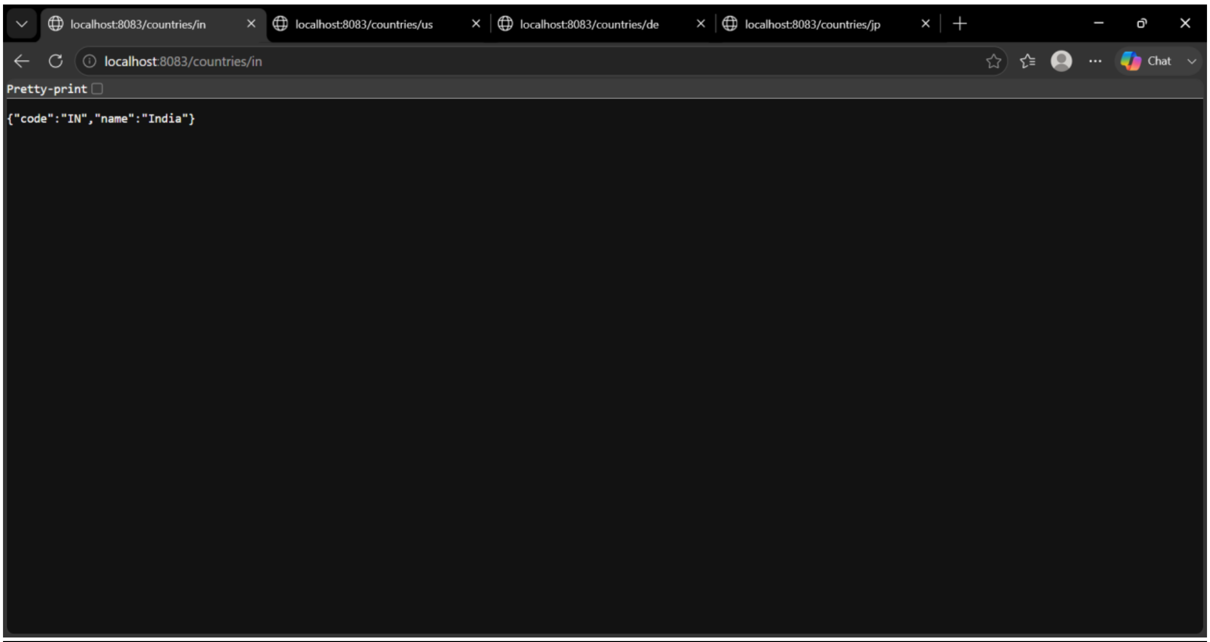


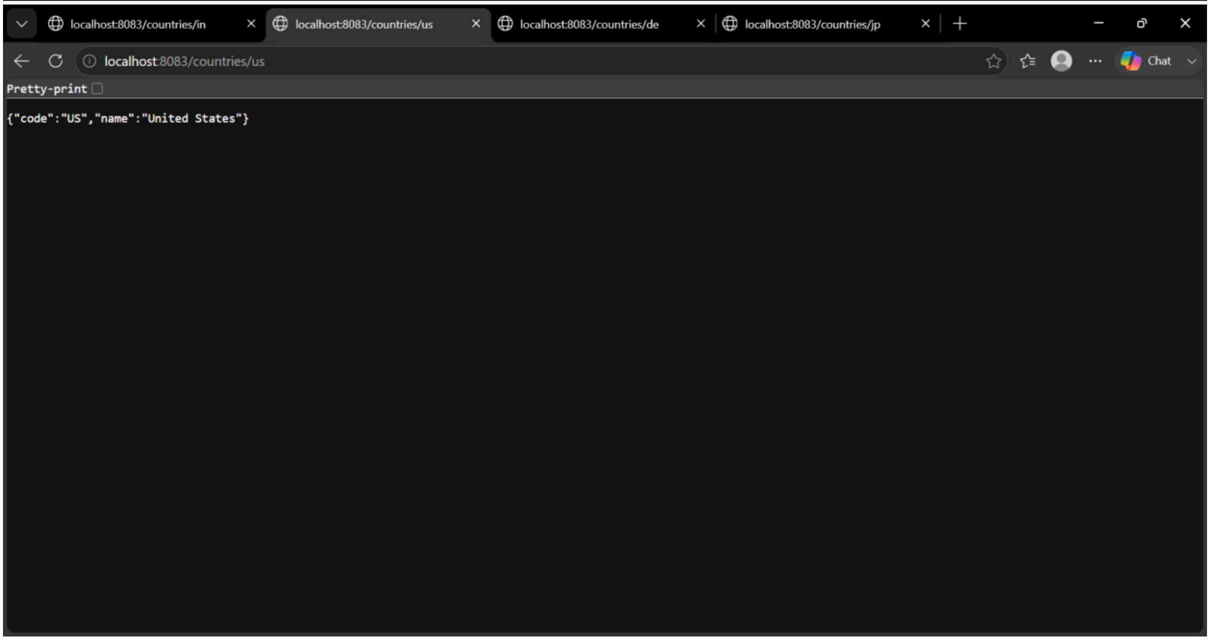
REST - Get country based on country code

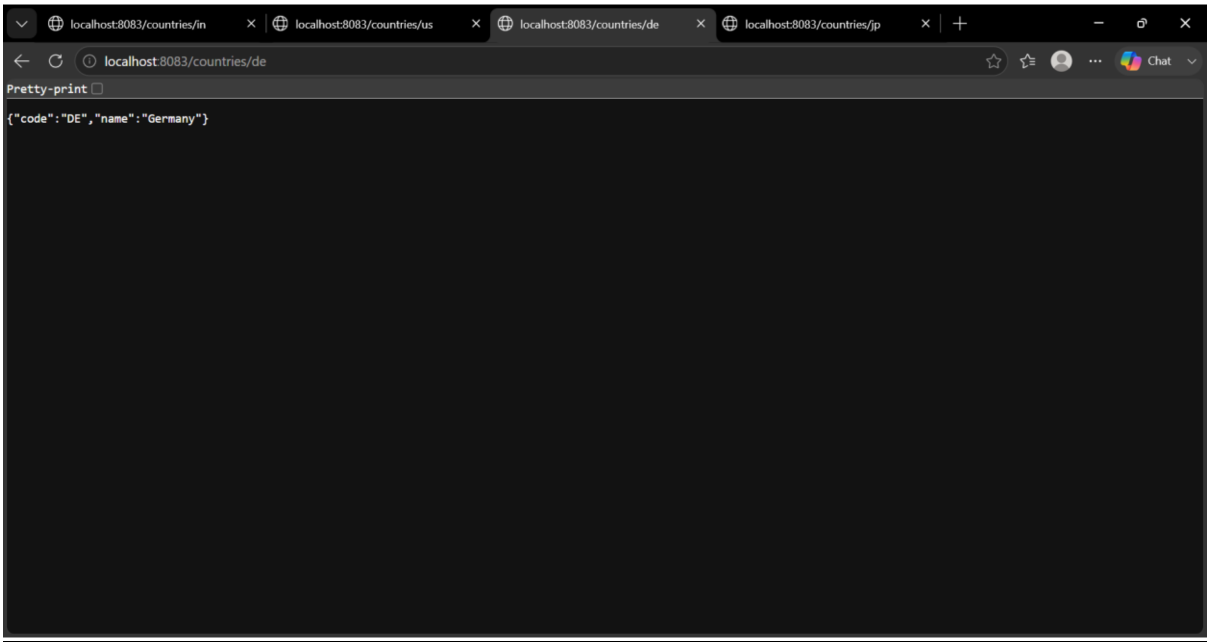


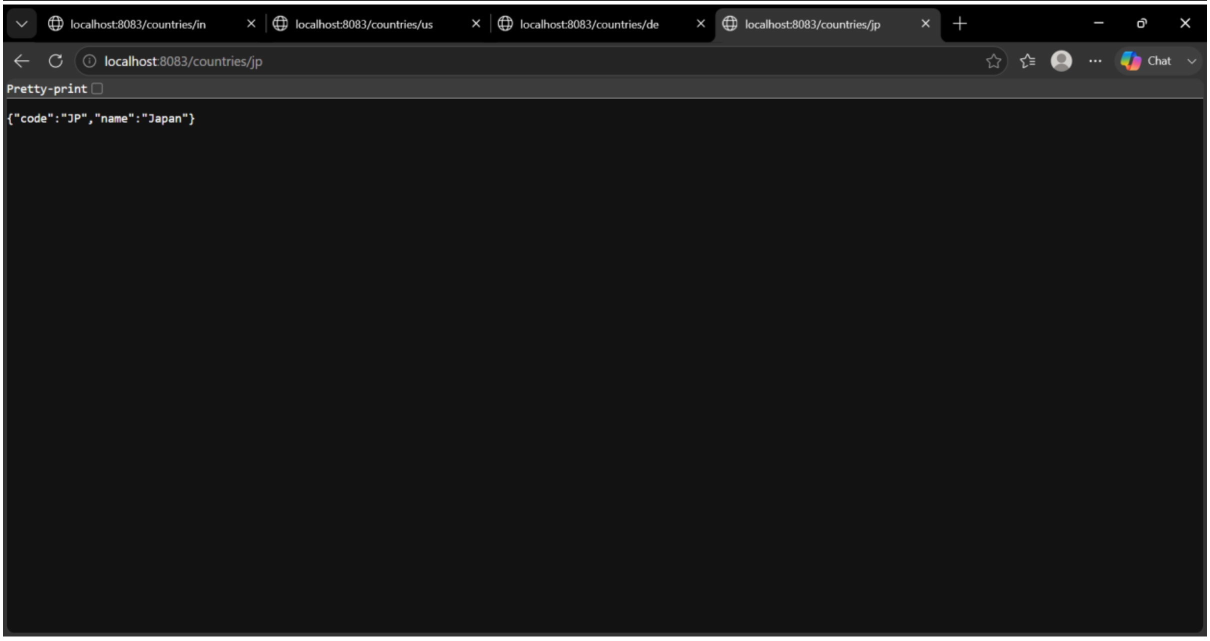

spring-learn
CountryService.java
src > main > java > com > cognizant > service > CountryService.java > CountryService
package com.cognizant.spring_learn.service;
import java.util.List;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.stereotype.Service;
import com.cognizant.spring_learn.Country;
@Service
public class CountryService {
@SuppressWarnings("unchecked")
public Country getCountry(String code) {
ApplicationContext context =
new ClassPathXmlApplicationContext("country.xml");
List<Country> countryList =
(List<Country>) context.getBean("countryList");
for (Country country : countryList) {
if (country.getCode().equalsIgnoreCase(code)) {
return country;
}
}
return null;
}
}


srini@KEERTHI-REDDY MINGW64 ~/Downloads/spring-learn/spring-learn
$ mvn spring-boot:run
2026-07-01T14:41:45.021+05:30 INFO 11804 --- [spring-learn] [restartedMain] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-01T14:41:45.972+05:30 INFO 11804 --- [spring-learn] [restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-07-01T14:41:46.341+05:30 WARN 11804 --- [spring-learn] [restartedMain] ConfigServletWebServerApplicationContext : Exception encountered during context initialization - cancelling refresh attempt: org.springframework.context.ApplicationContextException: Failed to start bean 'webServerStartStop'
Error starting ApplicationContext. To display the condition evaluation report re-run your application with 'debug' enabled.
APPLICATION FAILED TO START
Description:
Web server failed to start. Port 8083 was already in use.
Action:
Identify and stop the process that's listening on port 8083 or configure this application to listen on another port.
[INFO] BUILD SUCCESS
[INFO] Total time: 6.118 s
[INFO] Finished at: 2026-07-01T14:41:46+05:30

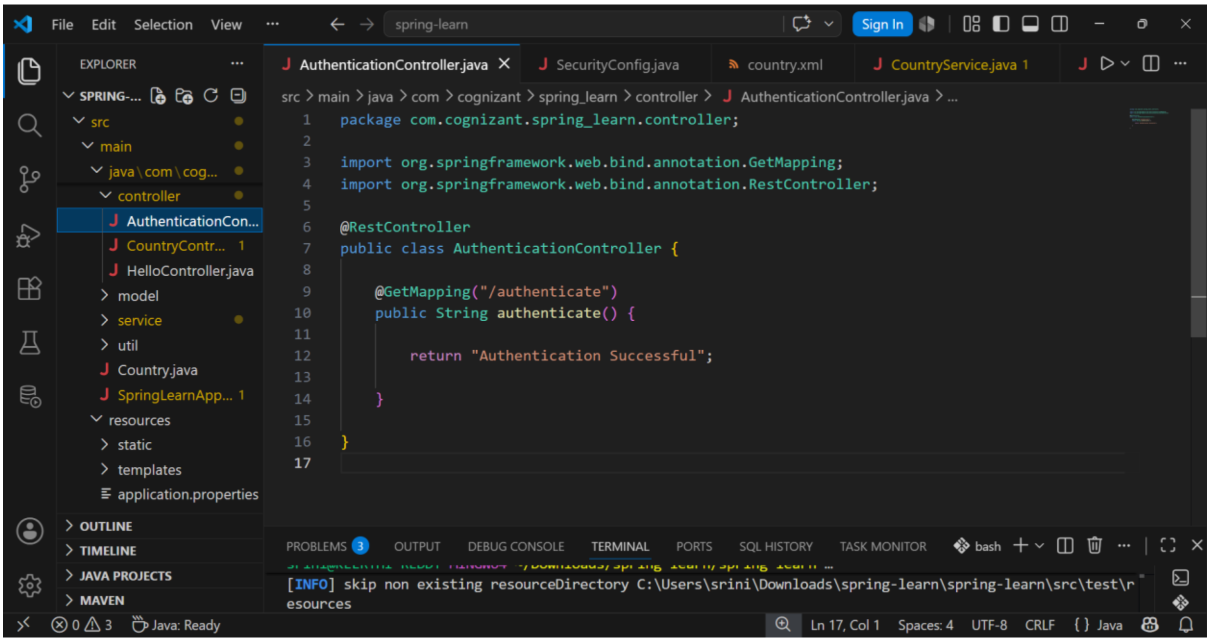

localhost:8083/countries/in
localhost:8083/countries/us
localhost:8083/countries/de
localhost:8083/countries/jp
localhost:8083/countries/in
Chat
Pretty-print
{"code":"IN","name":"India"}

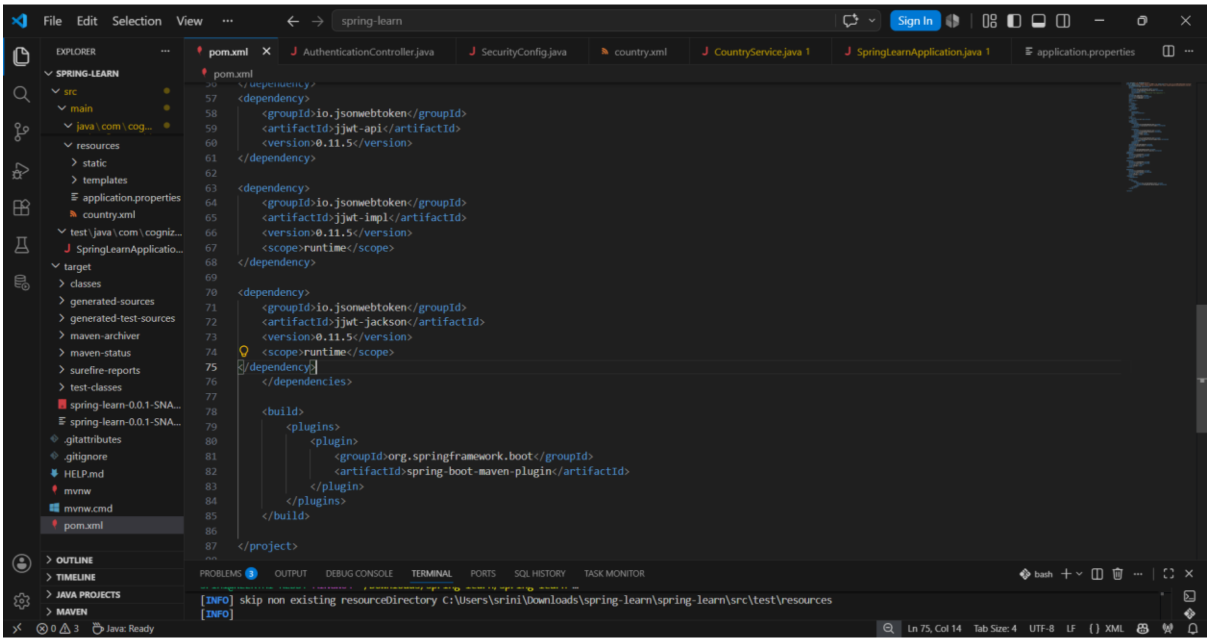

localhost:8083/countries/in
localhost:8083/countries/us
localhost:8083/countries/de
localhost:8083/countries/jp
localhost:8083/countries/us
Chat
Pretty-print
{"code":"US","name":"United States"}

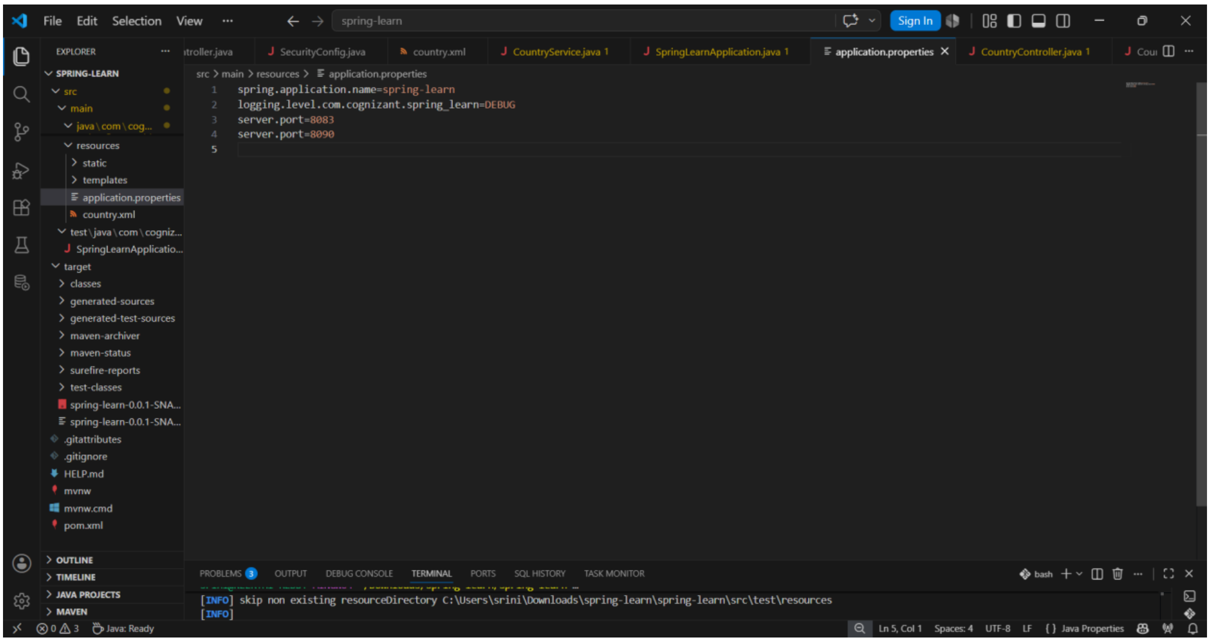

localhost:8083/countries/in
localhost:8083/countries/us
localhost:8083/countries/de
localhost:8083/countries/jp
localhost:8083/countries/de
Chat
Pretty-print
{"code":"DE","name":"Germany"}

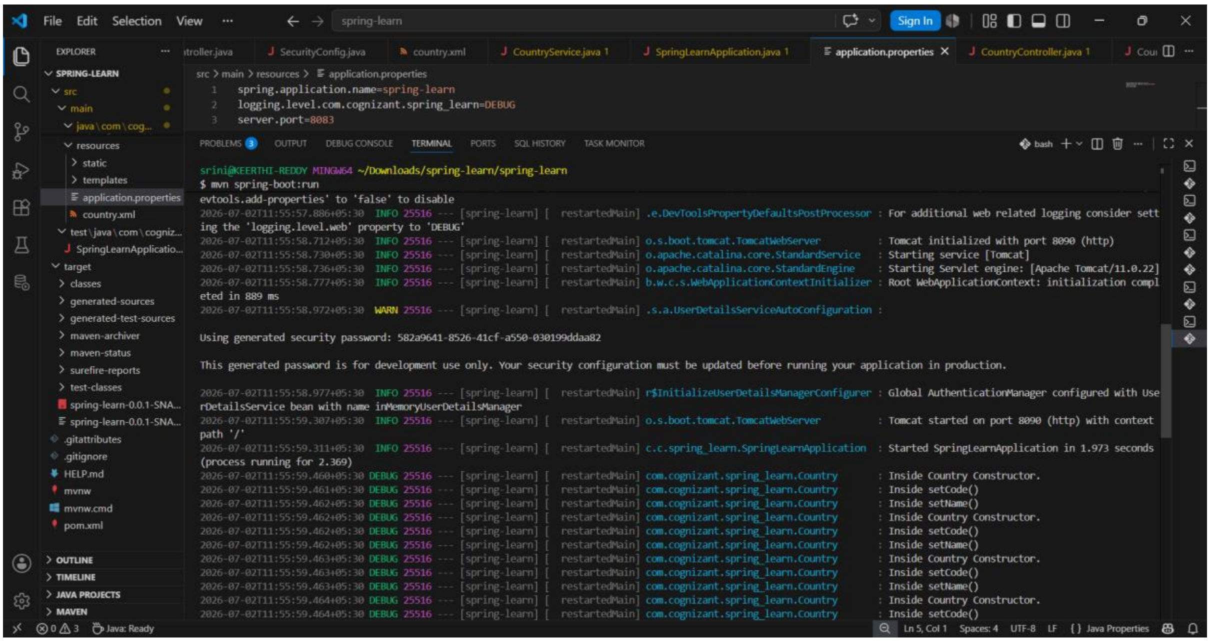

localhost:8083/countries/in
localhost:8083/countries/us
localhost:8083/countries/de
localhost:8083/countries/jp
localhost:8083/countries/jp
Chat
Pretty-print
{"code":"JP","name":"Japan"}

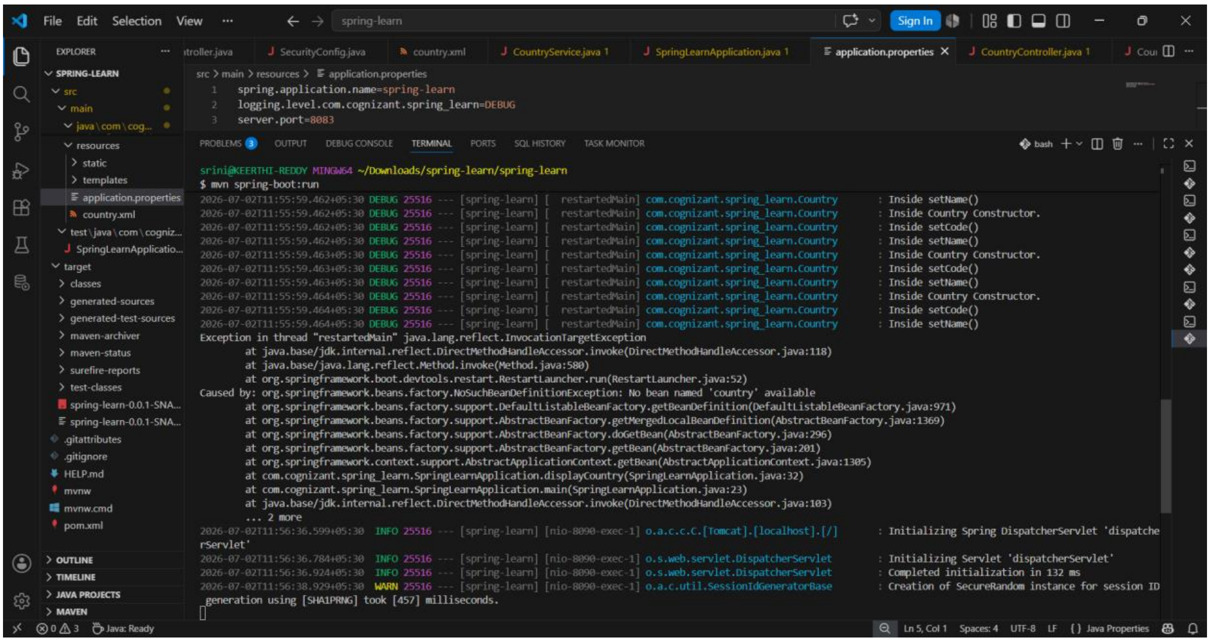
Create authentication service that returns JWT

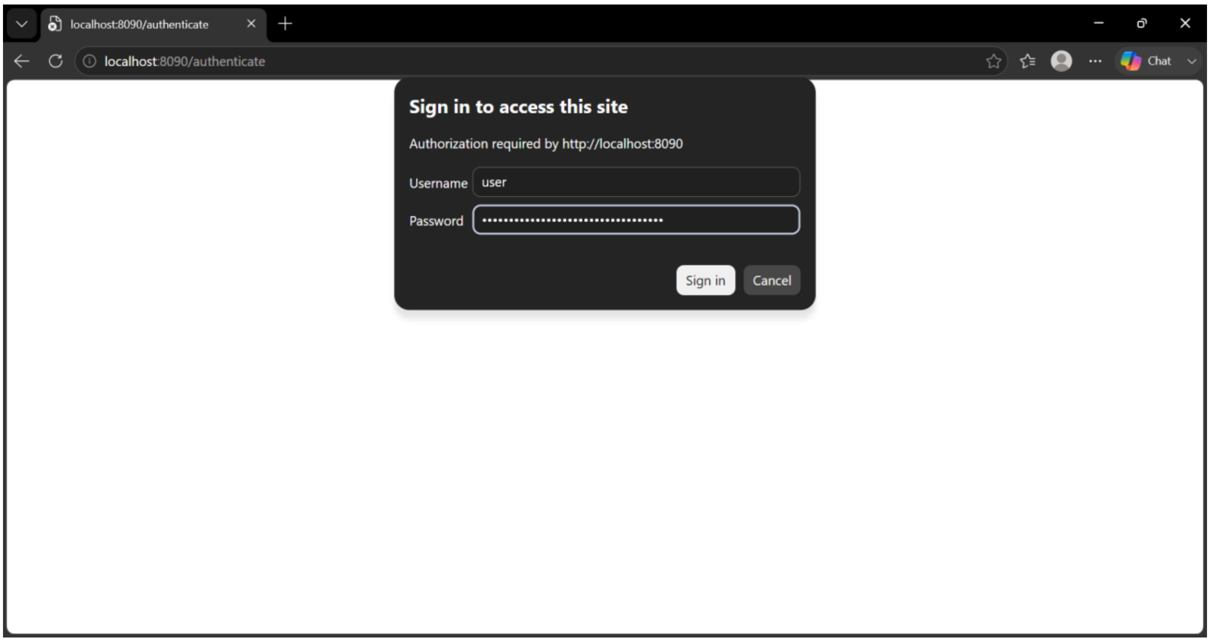





spring-learn
application.properties
spring.application.name=spring-learn
logging.level.com.cognizant.spring_learn=DEBUG
server.port=8083
server.port=8090
[INFO] skip non existing resourceDirectory C:\Users\srini\Downloads\spring-learn\spring-learn\src\test\resources
[INFO]


spring-learn
application.properties
spring.application.name=spring-learn
logging.level.com.cognizant.spring_learn=DEBUG
server.port=8083
srini@KEERTHI-REDDY MINGW64 ~/Downloads/spring-learn/spring-learn
$ mvn spring-boot:run
evtools.add-properties' to 'false' to disable
2026-07-02T11:55:57.886+05:30 INFO 25516 --- [spring-learn] [restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property to 'DEBUG'
2026-07-02T11:55:58.712+05:30 INFO 25516 --- [spring-learn] [restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8090 (http)
2026-07-02T11:55:58.730+05:30 INFO 25516 --- [spring-learn] [restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-02T11:55:58.736+05:30 INFO 25516 --- [spring-learn] [restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-02T11:55:58.777+05:30 INFO 25516 --- [spring-learn] [restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 889 ms
2026-07-02T11:55:58.972+05:30 WARN 25516 --- [spring-learn] [restartedMain] .s.a.UserDetailsServiceAutoConfiguration :
Using generated security password: 582a9641-8526-41cf-a550-030199ddaa82
This generated password is for development use only. Your security configuration must be updated before running your application in production.
2026-07-02T11:55:58.977+05:30 INFO 25516 --- [spring-learn] [restartedMain] r$InitializeUserDetailsManagerConfigurer : Global AuthenticationManager configured with UserDetailsService bean with name inMemoryUserDetailsManager
2026-07-02T11:55:59.307+05:30 INFO 25516 --- [spring-learn] [restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8090 (http) with context path '/'
2026-07-02T11:55:59.311+05:30 INFO 25516 --- [spring-learn] [restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 1.973 seconds (process running for 2.369)
2026-07-02T11:55:59.460+05:30 DEBUG 25516 --- [spring-learn] [restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-02T11:55:59.461+05:30 DEBUG 25516 --- [spring-learn] [restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-02T11:55:59.462+05:30 DEBUG 25516 --- [spring-learn] [restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-02T11:55:59.462+05:30 DEBUG 25516 --- [spring-learn] [restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-02T11:55:59.462+05:30 DEBUG 25516 --- [spring-learn] [restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-02T11:55:59.462+05:30 DEBUG 25516 --- [spring-learn] [restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-02T11:55:59.463+05:30 DEBUG 25516 --- [spring-learn] [restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-02T11:55:59.463+05:30 DEBUG 25516 --- [spring-learn] [restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-02T11:55:59.463+05:30 DEBUG 25516 --- [spring-learn] [restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-02T11:55:59.464+05:30 DEBUG 25516 --- [spring-learn] [restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-02T11:55:59.464+05:30 DEBUG 25516 --- [spring-learn] [restartedMain] com.cognizant.spring_learn.Country : Inside setCode()


File Edit Selection View
spring-learn
EXPLORER
SPRING-LEARN
src > main > resources > application.properties
spring.application.name=spring-learn
logging.level.com.cognizant.spring_learn=DEBUG
server.port=8083
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SQL HISTORY TASK MONITOR
srini@KEERTHI-REDDY MINGW64 ~/Downloads/spring-learn/spring-learn
$ mvn spring-boot:run
Exception in thread "restartedMain" java.lang.reflect.InvocationTargetException
Caused by: org.springframework.beans.factory.NoSuchBeanDefinitionException: No bean named 'country' available
Initializing Spring DispatcherServlet 'dispatcherServlet'
Initializing Servlet 'dispatcherServlet'
Completed initialization in 132 ms
Creation of SecureRandom instance for session ID generation using [SHA1PRNG] took [457] milliseconds.


localhost:8090/authenticate
Sign in to access this site
Authorization required by http://localhost:8090
Username
user
Password
Sign in
Cancel


localhost:8090/authenticate
localhost:8090/authenticate
Summarize
Authentication Successful